POLD1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes POLD1 gene fusions from the msk_impact_50k_2026 study, covering 16 fusion events. 2/6 fusions with known frame status (33.3%) were in-frame. 6/13 fusions (46.2%) retained the DNA polymerase family B domain. Domain retention was tested with Fisher's exact test (p=0.192308, not statistically significant at alpha=0.05). No literature benchmark is configured for POLD1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 16 analyzed fusion events, identifying 15 distinct fusion partner genes. The most frequent partner was MYH14, observed in 2 events.

Domain retention

DNA polymerase family B domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 2/2 (100.0%) of in-frame fusions retained the domain, p=0.192308 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.031968.

Domain disruption

Disruption of the DNA polymerase family B, exonuclease domain was tested with Fisher's exact test comparing in-frame fusions against all others; 1/2 (50.0%) of in-frame fusions disrupted the domain, p=0.871795 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.792079.

Junction concentration

22% of 9 fusions with this gene as the 3' partner join at exon 5, against 1% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.01998, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 13 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 415 aa (exon 11; label-permutation p=0.010989). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -62 aa from the nearest configured domain boundary at 477 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=6) and "not_retained" (n=7). For Frame_status, retained: 2/3 (66.7%, 95% CI 20.8%-93.9%); not_retained: 0/3 (0.0%, 95% CI 0.0%-56.1%). Welch's t-test comparing Tumor_variant_count between groups gave means of 24.8333 vs 43, p=0.299234 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score produced results for this run (15 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for POLD1 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (8 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 0 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: POLD1 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: POLD1 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (6 row(s) in window_scan, 4 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
MYH14	2	0.125	0.14952840991594127	0.010123138679069898	1.0	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2622894559031692	1
TMEM87A	1	0.0625	0.14952840991594127	0.010123138679069898	0.8578947368421053	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.22222366642948505	2
NR1H2	1	0.0625	0.14952840991594127	0.010123138679069898	0.7157894736842105	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2009078769558008	3
ZNF284	1	0.0625	0.14952840991594127	0.010123138679069898	0.6210526315789473	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1866973506400113	4
LILRB1	1	0.0625	0.14952840991594127	0.010123138679069898	0.5859649122807018	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1814341927452745	5
NDUFA3	1	0.0625	0.14952840991594127	0.010123138679069898	0.5701754385964912	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.17906577169264293	6
PNPLA6	1	0.0625	0.14952840991594127	0.010123138679069898	0.4666666666666667	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1635394559031692	7
PEPD	1	0.0625	0.14952840991594127	0.010123138679069898	0.3298245614035088	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.14301314011369554	8
SIGLECL1	1	0.0625	0.14952840991594127	0.010123138679069898	0.3070175438596491	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1395920874821166	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
C5AR1	1	0.0625	0.14952840991594127	0.010123138679069898	0.3035087719298246	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.13906577169264292	10
ADAMTSL1	1	0.0625	0.14952840991594127	0.010123138679069898	None	0.3538272568836993	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11004641870961086	11
CIC	1	0.0625	0.14952840991594127	0.010123138679069898	None	0.3538272568836993	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11004641870961086	12
SRRM1	1	0.0625	0.14952840991594127	0.010123138679069898	0.07719298245614037	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.10511840327159029	13
KLK10	1	0.0625	0.14952840991594127	0.010123138679069898	0.01754385964912286	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.09617103485053766	14
SPIB	1	0.0625	0.14952840991594127	0.010123138679069898	0.0	0.3538272568836993	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.09353945590316923	15

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
18	1	5	0	7	None	0.46153846153846156	0.3357921019231931
170	2	4	0	7	None	0.1923076923076923	0.7160033436347992
179	3	3	0	7	None	0.06993006993006992	1.1553360374650619
253	4	2	0	7	None	0.02097902097902098	1.6782147827453993
415	5	1	0	7	None	0.004662004662004663	2.3314272965207428
496	5	1	1	6	30.0	0.029137529137529136	1.5355472791766678
631	5	1	2	5	12.5	0.10256410256410255	0.9890046156985369
719	5	1	3	4	6.666666666666667	0.26573426573426573	0.5755524408482516
797	5	1	4	3	3.75	0.5594405594405595	0.2522460504731182
810	5	1	5	2	2.0	1.0	-0.0
941	5	1	6	1	0.8333333333333334	1.0	-0.0

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
975	5	1	7	0	0.0	0.46153846153846156	0.3357921019231931

domain_disruption tables

frame_domain_contingency_table

1	7
1	4

frame_domain_contingency_tables_by_tumor_type

stratum	table
COAD	[0, 0], [0, 1]
CUP	[0, 0], [1, 0]
GB	[0, 0], [0, 1]
GEJ	[0, 1], [0, 0]
IDC	[0, 2], [0, 0]
LUAD	[0, 0], [0, 0]
OS	[1, 0], [0, 0]
PAAD	[0, 1], [0, 1]
SCLC	[0, 0], [0, 1]
SKCM	[0, 1], [0, 0]
UCCC	[0, 1], [0, 0]
UEC	[0, 1], [0, 0]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf03104	DNA polymerase family B, exonuclease domain	13	5	4	4	0.3224063400576369

domain_retention tables

frame_domain_contingency_table

2	4
0	7

frame_domain_contingency_tables_by_tumor_type

stratum	table
COAD	[0, 0], [0, 1]
CUP	[1, 0], [0, 0]
GB	[0, 0], [0, 1]
GEJ	[0, 1], [0, 0]
IDC	[0, 0], [0, 2]
OS	[1, 0], [0, 0]
PAAD	[0, 1], [0, 1]
SCLC	[0, 1], [0, 0]
SKCM	[0, 0], [0, 1]
UCCC	[0, 1], [0, 0]
UEC	[0, 0], [0, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	DNA polymerase family B	13	6	6	1	0.2753912753912754

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
13	8	16	0.5

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ADAMTSL1	1	1	1	1
C5AR1	1	1	1	1
CIC	1	1	1	1
KLK10	1	1	1	1
LILRB1	1	1	1	1
MYH14	2	2	2	2
NDUFA3	1	1	1	1

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
NR1H2	1	1	1	1
PEPD	1	1	1	1
PNPLA6	1	1	1	1
SIGLECL1	1	1	1	1
SPIB	1	1	1	1
SRRM1	1	1	1	1
TMEM87A	1	1	1	1
ZNF284	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
19	50905000	50906000	3	EVT-P-0003038-T01-IM5-10, EVT-P-0016377-T01-IM6-11, EVT-P-0051640-T01-IM6-5
19	50919000	50920000	3	EVT-P-0014626-T01-IM6-34, EVT-P-0032778-T01-IM6-9, EVT-P-0047491-T01-IM6-33
19	50909000	50910000	2	EVT-P-0019604-T01-IM6-4, EVT-P-0064872-T01-IM7-35
19	50902000	50903000	1	EVT-P-0069297-T01-IM7-1
19	50912000	50913000	1	EVT-P-0017653-T01-IM5-36
19	50916000	50917000	1	EVT-P-0005493-T01-IM5-7
19	50917000	50918000	1	EVT-P-0052327-T01-IM6-6
19	50918000	50919000	1	EVT-P-0023588-T01-IM6-32

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	5	2	9	0.2222222222222222	0.005938947618482005	2.9083208738731137
three_prime	23	2	9	0.2222222222222222	0.020845706140871837	1.8479812906852997
five_prime	12	1	4	0.25	0.017104169141228174	1.1759960015198307
five_prime	15	1	4	0.25	0.010007126737142179	1.4041550500708753
five_prime	20	1	4	0.25	0.018499821831571447	1.1428422486509395

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	23	1	4	0.25	0.006354673951775745	1.598988614112688
three_prime	2	1	9	0.1111111111111111	0.43695807102981354	0.0024767875326987335
three_prime	7	1	9	0.1111111111111111	0.004810547570970424	1.3719062570618847
three_prime	11	1	9	0.1111111111111111	0.08091816130181732	0.2740371815060575
three_prime	18	1	9	0.1111111111111111	0.11447321534624065	0.17706279286066873
three_prime	20	1	9	0.1111111111111111	0.03299085402066754	0.5840136402076834

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
610	810	200	4	9	0	6	0.0	0.06993006993006994	1.1553360374650619	EVT-P-0005493-T01-IM5-7, EVT-P-0017653-T01-IM5-36, EVT-P-0023588-T01-IM6-32, EVT-P-0052327-T01-IM6-6
631	831	200	4	9	0	6	0.0	0.06993006993006994	1.1553360374650619	EVT-P-0005493-T01-IM5-7, EVT-P-0017653-T01-IM5-36, EVT-P-0023588-T01-IM6-32, EVT-P-0052327-T01-IM6-6
775	975	200	4	9	0	6	0.0	0.06993006993006994	1.1553360374650619	EVT-P-0014626-T01-IM6-34, EVT-P-0023588-T01-IM6-32, EVT-P-0032778-T01-IM6-9, EVT-P-0052327-T01-IM6-6
785	985	200	5	8	1	5	0.15	0.26573426573426573	0.5755524408482516	EVT-P-0014626-T01-IM6-34, EVT-P-0023588-T01-IM6-32, EVT-P-0032778-T01-IM6-9, EVT-P-0047491-T01-IM6-33, EVT-P-0052327-T01-IM6-6
797	997	200	5	8	1	5	0.15	0.26573426573426573	0.5755524408482516	EVT-P-0014626-T01-IM6-34, EVT-P-0023588-T01-IM6-32, EVT-P-0032778-T01-IM6-9, EVT-P-0047491-T01-IM6-33, EVT-P-0052327-T01-IM6-6
810	1010	200	4	9	1	5	0.26666666666666666	0.5594405594405595	0.2522460504731182	EVT-P-0014626-T01-IM6-34, EVT-P-0023588-T01-IM6-32, EVT-P-0032778-T01-IM6-9, EVT-P-0047491-T01-IM6-33

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
610	810	200	4	9	0	6	0.0	0.06993006993006994	1.1553360374650619	EVT-P-0005493-T01-IM5-7, EVT-P-0017653-T01-IM5-36, EVT-P-0023588-T01-IM6-32, EVT-P-0052327-T01-IM6-6

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
775	975	200	4	9	0	6	0.0	0.06993006993006994	1.1553360374650619	EVT-P-0014626-T01-IM6-34, EVT-P-0023588-T01-IM6-32, EVT-P-0032778-T01-IM6-9, EVT-P-0052327-T01-IM6-6
785	985	200	5	8	1	5	0.15	0.26573426573426573	0.5755524408482516	EVT-P-0014626-T01-IM6-34, EVT-P-0023588-T01-IM6-32, EVT-P-0032778-T01-IM6-9, EVT-P-0047491-T01-IM6-33, EVT-P-0052327-T01-IM6-6
810	1010	200	4	9	1	5	0.26666666666666666	0.5594405594405595	0.2522460504731182	EVT-P-0014626-T01-IM6-34, EVT-P-0023588-T01-IM6-32, EVT-P-0032778-T01-IM6-9, EVT-P-0047491-T01-IM6-33

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006929 7-T01-IM7-1	P-006929 7-T01-IM7	P-006929 7	Small Cell Lung Cancer	SCLC	C5AR1-POLD1	C5AR1	unknown	three_prime	50902161	2	18	False	18	True	exact match	retained	1.0	False													DNA polymerase family B; DNA polymerase family B, exonuclease domain			C5AR1 (NM_001736) - POLD1 (NM_002691) rearrangement: c.3+879:C5AR1_c.53:POLD1del | Protein Fusion: mid-exon {C5AR1:POLD1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001960 4-T01-IM6-4	P-001960 4-T01-IM6	P-001960 4	Bone Cancer	OS	MYH14-POLD1	MYH14	in-frame	three_prime	50909319	11	415	True	415	True	exact match	retained	1.0	False													DNA polymerase family B		DNA polymerase family B, exonuclease domain	MYH14 (NM_001145809) - POLD1 (NM_002691) Rearrangement : c.4539+61:MYH14_c.1243-120:POLD1del | Protein Fusion: in frame {MYH14:POLD1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005164 0-T01-IM6-5	P-005164 0-T01-IM6	P-005164 0	Esophagogastric Cancer	GEJ	NR1H2-POLD1	NR1H2	out-of-frame	three_prime	50905674	7	253	True	253	True	exact match	retained	1.0	False													DNA polymerase family B, exonuclease domain		DNA polymerase family B, exonuclease domain	NR1H2 (NM_007121) - POLD1 (NM_002691) rearrangement: c.1028-692:NR1H2_c.759-37:POLD1del | Protein Fusion: out of frame {NR1H2:POLD1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005232 7-T01-IM6-6	P-005232 7-T01-IM6	P-005232 7	Breast Cancer	IDC	PEPD-POLD1	PEPD	out-of-frame	three_prime	50917974	20	797	True	797	True	exact match	disrupted	0.424242424242425	True	partner_lacks_domain	PF00136					323			False			DNA polymerase family B, exonuclease domain	DNA polymerase family B	PEPD (NM_000285) - POLD1 (NM_002691) rearrangement: c.967+204:PEPD_c.2389-98:POLD1inv | Protein Fusion: out of frame {PEPD:POLD1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000549 3-T01-IM5-7	P-000549 3-T01-IM5	P-000549 3	Melanoma	SKCM	PNPLA6-POLD1	PNPLA6	unknown	three_prime	50916627	18	719	True	719	True	exact match	disrupted	0.606060606060601	True	partner_lacks_domain	PF00136					441			True			DNA polymerase family B, exonuclease domain	DNA polymerase family B	PNPLA6 (NM_001166111) - POLD1 (NM_002691) rearrangement: c.1323:PNPLA6_c.2155-56:POLD1del | Protein fusion: mid-exon (PNPLA6-POLD1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003277 8-T01-IM6-9	P-003277 8	P-003277 8	Endometrial Cancer	UEC	POLD1-KLK10	KLK10	unknown	three_prime	50919757	23	975	False	975	True	exact match	disrupted	0.009324009324	True	partner_lacks_domain	PF00136					30			False			DNA polymerase family B, exonuclease domain	DNA polymerase family B	KLK10 (NM_001077500) - POLD1 (NM_002691) Rearrangement : c.88+823:KLK10_c.2925:POLD1inv | Protein Fusion : mid-exon {KLK10:POLD1}	NA
EVT-P-000303 8-T01-IM5-10	P-000303 8	P-000303 8	Endometrial Cancer	UCCC	POLD1-LILRB1	LILRB1	unknown	three_prime	50905327	5	179	False	179	True	exact match	retained	1.0	False												DNA polymerase family B		DNA polymerase family B, exonuclease domain	NA | Protein fusion: mid-exon (LILRB1-POLD1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001637-T01-IM6-11	P-001637-T01-IM6	P-001637	Pancreatic Cancer	PAAD	POLD1-NDUFA3	NDUFA3	unknown	three_prime	50905302	5	170	False	170	True	exact match	retained	1.0	False													DNA polymerase family B		DNA polymerase family B, exonuclease domain	NDUFA3 (NM_004542) - POLD1 (NM_002691) rearrangement: c.85+584:NDUFA3_c.510:POLD1dup | Protein Fusion: mid-exon {NDUFA3:POLD1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002358-T01-IM6-32	P-002358-T01-IM6	P-002358	Colorectal Cancer	COAD	POLD1-SIGLECL1	SIGLECL1	unknown	five_prime	50918112	20	810	False	809	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.6083916083916084	True	partner_lacks_domain	PF00136					1			False		DNApolymerase family B, exonuclease domain		DNApolymerase family B	POLD1 (NM_002691) - SIGLECL1 (NM_173635) Rearrangement : c.2429:POLD1_c.-191+1669:SIGLECL1del | Protein Fusion: mid-exon {POLD1:SIGLECL1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004749 1-T01-IM6-33	P-004749 1-T01-IM6	P-004749 1	Cancer of Unknown Primary	CUP	POLD1-SPIB	SPIB	in-frame	five_prime	50919861	23	985	True	984	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False														DNA polymerase family B; DNA polymerase family B, exonuclease domain				POLD1 (NM_002691) - SPIB (NM_003121) rearrangement: c.2954-6:POLD1_c.490+532:SPIBdel | ProteinFusion: in frame {POLD1:SPIB}	NA

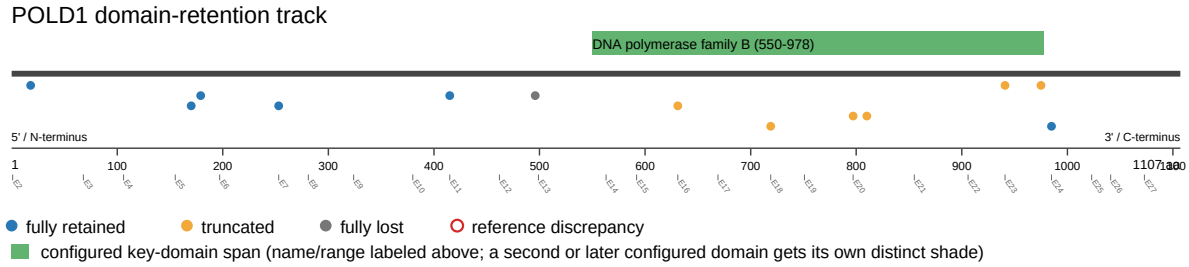
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001462 6-T01-IM6-34	P-001462 6-T01-IM6	P-001462 6	Breast Cancer	IDC	POLD1-SRRM1	SRRM1	out-of-frame	three_prime	50919623	23	941	True	941	True	exact match	disrupted	0.088578088578088	True	partner_lacks_domain	PF00136					347			False			DNA polymerase family B, exonuclease domain	DNA polymerase family B	SRRM1 (NM_005839) - POLD1 (NM_002691) rearrangement: t(1;19)(p36.11;q13.33)(chr1:g.24979609::chr19:g.50919623) | Protein fusion: out of frame (SRRM1-POLD1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006487 2-T01-IM7 -35	P-006487 2-T01-IM7	P-006487 2	Glioma	GB	POLD1-TMEM87A	TMEM87A	unknown	five_prime	50909766	12	496	False	495	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False															DNA polymerase family B, exonuclease domain	DNA polymerase family B		POLD1 (NM_002691) - TMEM87A (NM_015497) rearrangement: t(15;19)(q15.1;q13.33)(chr15:g.42511439::chr19:g.50909766)| Protein Fusion: mid-exon { POLD1:TMEM87A}	NA

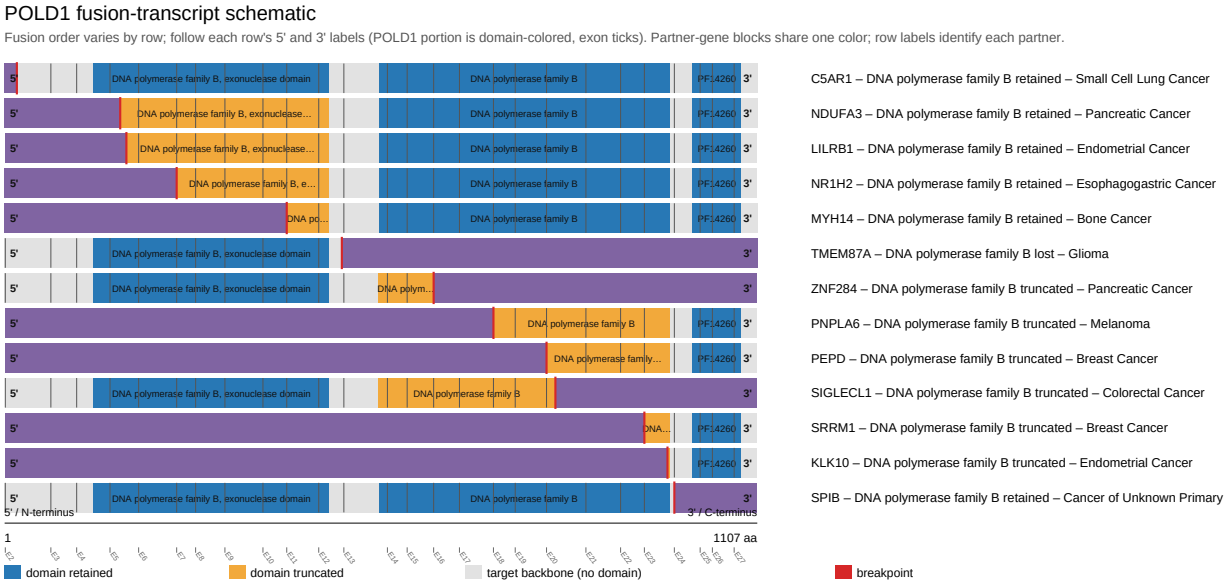
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001765 3-T01-IM5-36	P-001765 3-T01-IM5	P-001765 3	Pancreatic Cancer	PAAD	ZNF284-POLD1	ZNF284	out-of-frame	five_prime	50912201	15	631	True	630	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.19114219114219114	True	partner_lacks_domain	PF00136						6			False		DNA polymerase family B, exonuclease domain		DNA polymerase family B	POLD1 (NM_002691) - ZNF284 (NM_001037813) rearrangement: c.1892+43:POLD1_c.15+1157:ZNF284dup | Protein Fusion: out of frame {POLD1:ZNF284}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.

intragenic deletion schematic

